

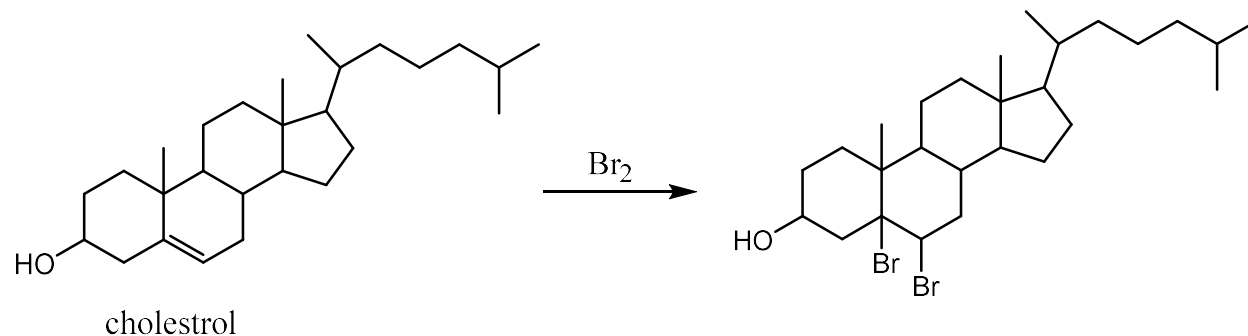
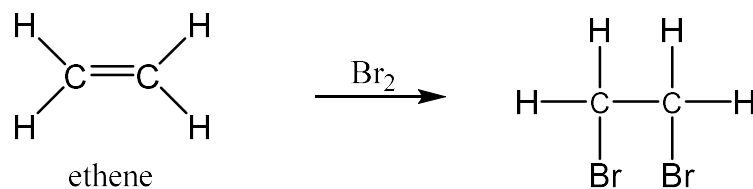
UNIT 2

FUNCTIONAL GROUPS IN ORGANIC CHEMISTRY

2.1 The importance of classification of organic compounds according to their functionality

A **functional group** is a particular combination of atoms in a molecule that is particularly responsible for characteristic set of reactions that the molecule exhibit.

- The chemistry of every organic molecule, regardless of size and complexity, is determined by the functional group it contains.



Unit 2

Nomenclature of organic compounds

- Common name

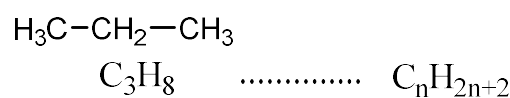
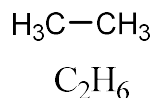
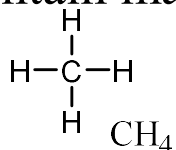
n-butane, isobutane, n-pentane, isopentane, and neopentane

- Systematic name

IUPAC rules

A. Nomenclature of ALKANE

- Alkanes are saturated hydrocarbons with C-C single bond. They are saturated, because contain maximum possible number of H per C.



Molecular Formula	Structural Formula	IUPAC Name
CH ₄	CH ₄	Methane
C ₂ H ₆	CH ₃ -CH ₃	Ethane
C ₃ H ₈	CH ₃ -CH ₂ -CH ₃	Propane
C ₄ H ₁₀	CH ₃ -(CH ₂) ₂ -CH ₃	Butane
C ₅ H ₁₂	CH ₃ -(CH ₂) ₃ -CH ₃	Pentane
C ₆ H ₁₄	CH ₃ -(CH ₂) ₄ -CH ₃	Hexane
C ₇ H ₁₆	CH ₃ -(CH ₂) ₅ -CH ₃	Heptane
C ₈ H ₁₈	CH ₃ -(CH ₂) ₆ -CH ₃	Octane
C ₉ H ₂₀	CH ₃ -(CH ₂) ₇ -CH ₃	Nonane
C ₁₀ H ₂₂	CH ₃ -(CH ₂) ₈ -CH ₃	Decane

Molecular Formula	Structural Formula	IUPAC Name
C ₁₁ H ₂₄	CH ₃ -(CH ₂) ₉ -CH ₃	Undecane
C ₁₂ H ₂₆	CH ₃ -(CH ₂) ₁₀ -CH ₃	Dodecane
C ₁₃ H ₂₈	CH ₃ -(CH ₂) ₁₁ -CH ₃	Tridecane
C ₁₄ H ₃₀	CH ₃ -(CH ₂) ₁₂ -CH ₃	Tetradecane
C ₁₅ H ₃₂	CH ₃ -(CH ₂) ₁₃ -CH ₃	Pentadecane
C ₂₀ H ₄₂	CH ₃ -(CH ₂) ₁₈ -CH ₃	Icosane
C ₃₀ H ₆₂		Triacontane
C ₄₀ H ₈₂		Tetracontane
C ₅₀ H ₁₀₂		Tentacontane
C ₁₀₀ H ₂₀₂		Hectane

IUPAC rule

IUPAC name typically has four parts: prefix, locant, parent, and suffix.

Prefix — Locant — Parent — Suffix

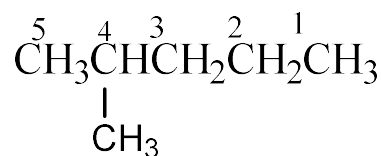
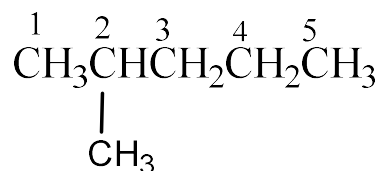
where and what are the substituents?

where is the primary functional group?

How many
carbon

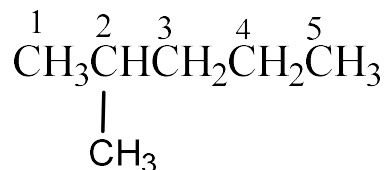
what is the primary functional group?

1. Find the parent hydrocarbon-Pick out the longest continuous carbon chain.
2. Number the longest continuous chain in the direction that gives the lowest number to the substituent group at the first point of branching.



incorrect

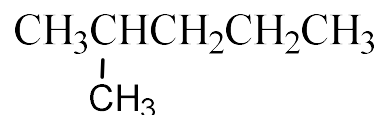
3. Identify and number the substituents.



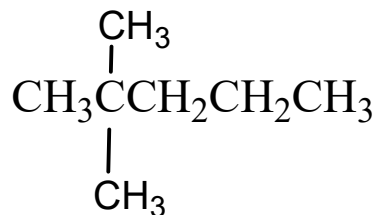
parent name: pentane

substituent: methyl (CH₃) on C-4

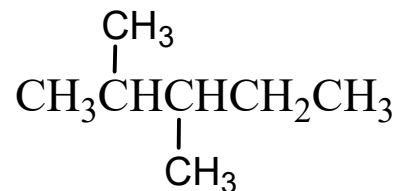
4. Write the name as a single word- use hyphens to separate the locants from words.



IUPAC name: 2-methylpentane



2,2-dimethylpentane

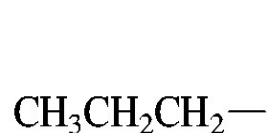
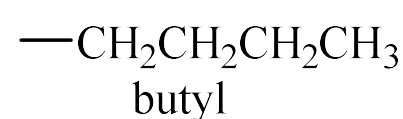
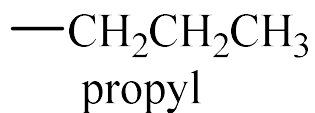
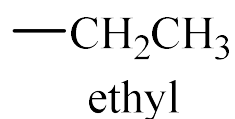
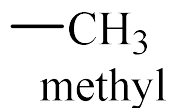


2,3-dimethylpentane

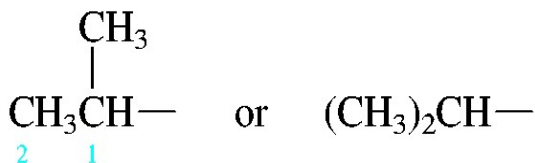
ALKYL GROUPS

An alkyl group lacks one of the hydrogens of an alkane.

Replacing the -ane endings -yl.



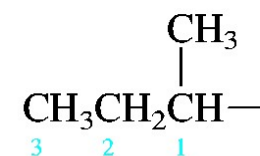
Propyl group
(common name: *n*-propyl)



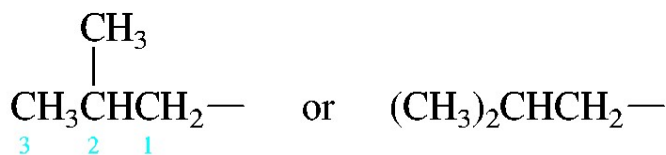
1-Methylethyl group
(common name: isopropyl)



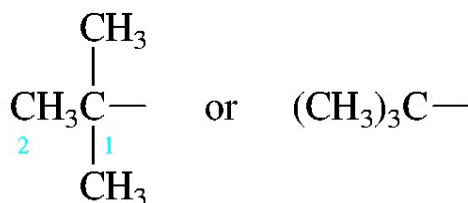
Butyl group
(common name: *n*-butyl)



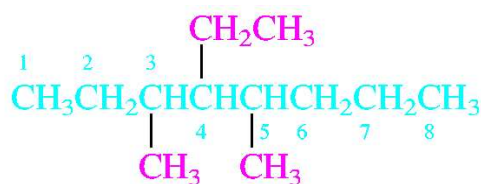
1-Methylpropyl group
(common name: *sec*-butyl)



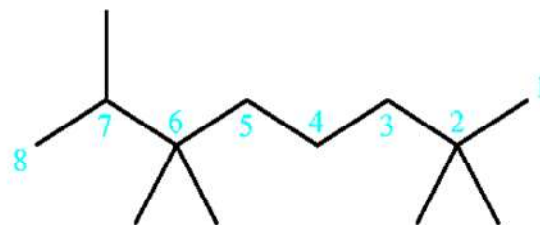
2-Methylpropyl group
(common name: isobutyl)



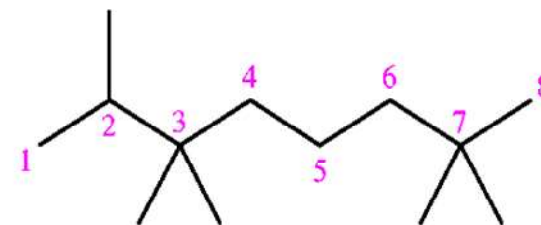
1,1-Dimethylethyl group
(common name: *tert*-butyl)



4-ethyl-3,5-dimethyloctane.



2,2,6,6,7-Pentamethyloctane
(correct)



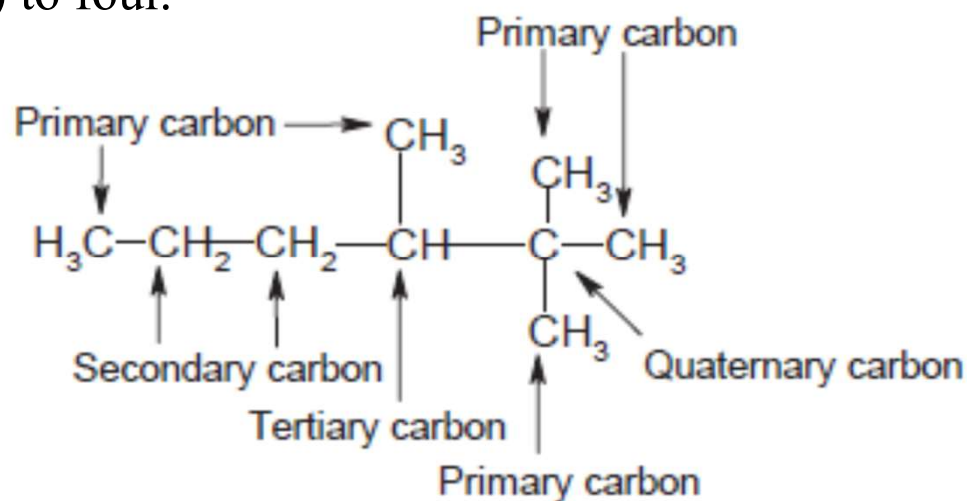
2,3,3,7,7-Pentamethyloctane
(incorrect!)

"first point of difference" rule

- Locants are not added together, but examined one by one.

Classification of carbon

- **Primary** carbon (1°) is directly attached to one other carbon.
- **Secondary** carbon (2°) is directly attached to two other carbons
- **Tertiary** carbon (3°) to three
- **Quaternary** carbon (4°) to four.



NOMENCLATURE OF CYCLOALKANE

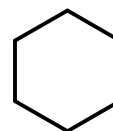
- prefix cyclo- to the name of the unbranched alkane with the same number of carbons as the ring.



cyclopropane

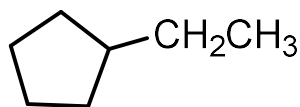


cyclopentane

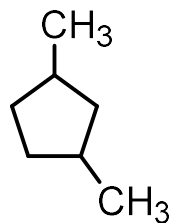


cyclohexane

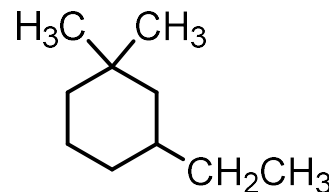
- Choose a point of attachment at C-1 and number the ring in the direction that gives the lowest number to the substituents at the first point of difference



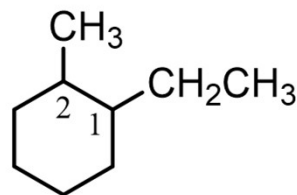
Ethylcyclopentane



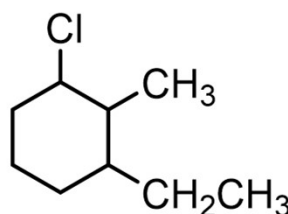
1,3 dimethylcyclopentane



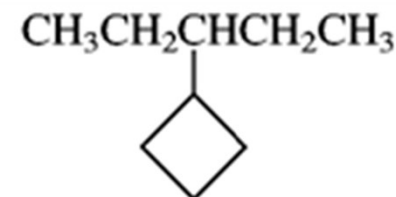
3-Ethyl-1,1-dimethylcyclohexane
not 1-Ethyl-3,3-dimethylcyclohexane



1-Ethyl-2-methylcyclohexane
not 2-Ethyl-1-methylcyclohexane



1-chloro-3-ethyl-2-methylcyclohexane



3-Cyclobutylpentane

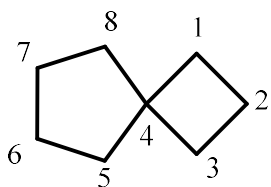
POLYCYCLIC ALKANES

a) Spirocyclic compounds: two rings share one common atom.

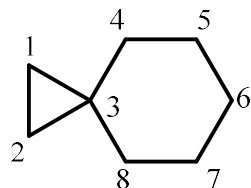
Named as spiro[n,m]alkane. n and m are number of atoms in each bridge (excluding the spiro atom)

Parent name is total number of carbon in the ring.

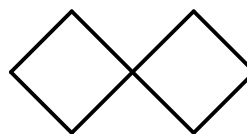
The c-atom in the smaller ring containing the common atom (spiro atom) is assign no. 1



Spiro[3,4]octane

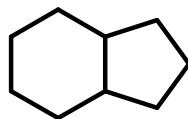
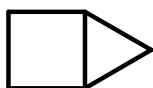


Spiro[2,5]octane



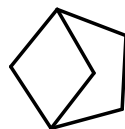
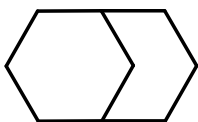
Spiro[3,3]heptane

b) Fused ring compounds: two rings share one common side



c) Bridge-ring compounds: two rings share more than one common side

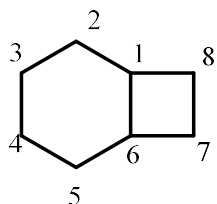
There are more than two common atoms which are adjacent to each other.



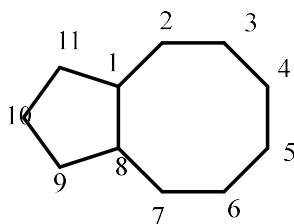
Both Fused ring and Bridge- ring compounds named using the same set of rules.

Use prefix bicyclo-

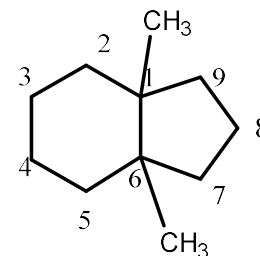
- Number the compound by beginning at one bridgehead, then proceeding around the 2nd longest and then finally to the shortest bridge.
- The number of atoms in each bridge placed in descending order, within brackets.



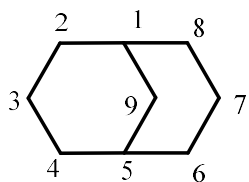
Bicyclo[4,2,0]octane



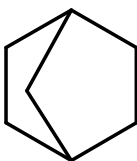
Bicyclo[6,3,0]undecane



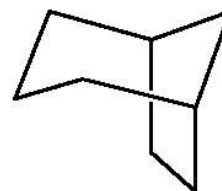
1,6-dimethylbicyclo[4,3,0]nonane



Bicyclo[3,3,1]nonane



Bicyclo[2,2,1]heptane



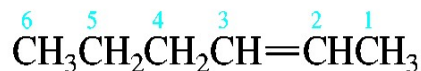
Bicyclo[3,2,1]octane

Nomenclature of ALKENES

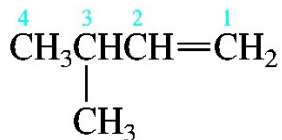
- Suffix : -ene
- The longest continuous includes the double bond.
- Indicate the position of the double bond, lower number



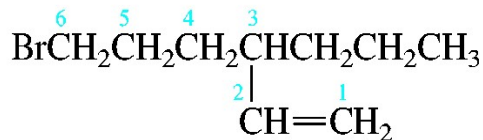
1-Butene
(not 1,2-butene)



2-Hexene
(not 4-hexene)



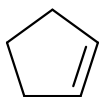
3-Methyl-1-butene
(not 2-methyl-3-butene)



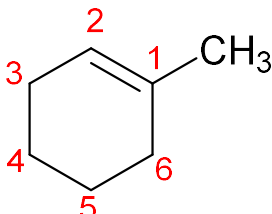
6-Bromo-3-propyl-1-hexene
(longest chain that contains double bond is six carbons)

Cycloalkenes

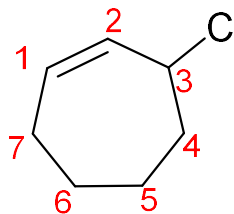
- Prefix –cyclo
- Number the cycloalkane so that the double bond is between C-1 and C-2.



Cyclopentene



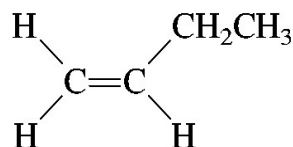
1-Methylcyclohexene



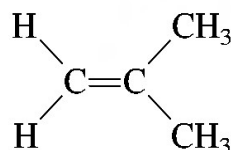
3-Chlorocycloheptene
not (1-chloro-2-cycloheptene)

ISOMERISM IN ALKENES

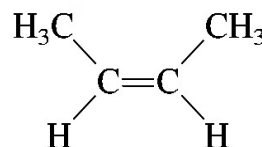
Alkenes show **cis-trans** isomerism.



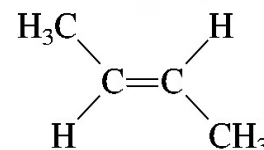
1-Butene



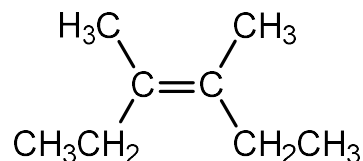
2-Methylpropene



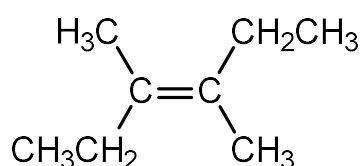
cis-2-Butene



trans-2-Butene

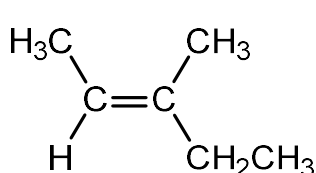
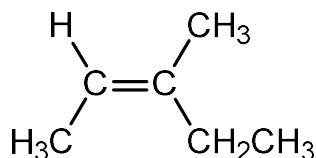


cis-3,4-dimethyl-3-hexene



trans-3,4-dimethyl-3-hexene

- ***cis*-isomer**: identical groups on the same side of the double bond
- ***trans*-isomer**: identical groups on opposite sides of the double bond.



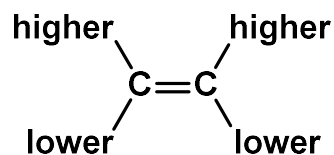
Cis or trans ?

Such isomers named based on an atomic number criterion for ranking substituents on the doubly bonded carbons. The system is called the *E/Z* nomenclature.

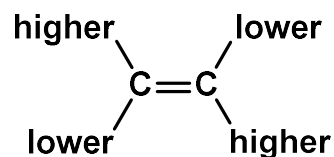
E/Z nomenclature

Cahn-Ingold-Prelog rules for assigning priority

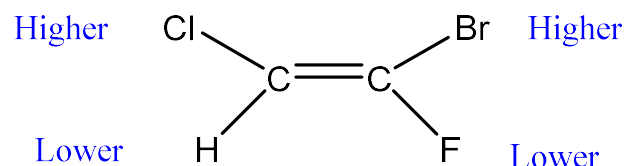
1. Look at atoms directly attached to the double bond. **Higher atomic number = higher priority**
2. If no difference exist at the first attachment atom, keep going down the line until the first difference reached.
3. Multiple bonded atoms are equivalent to the same number of single bonded atoms.



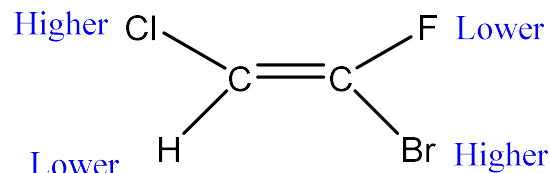
Z (zusammen)



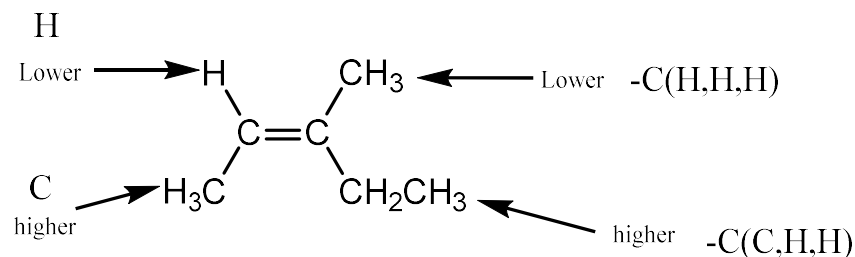
E (entgegen)



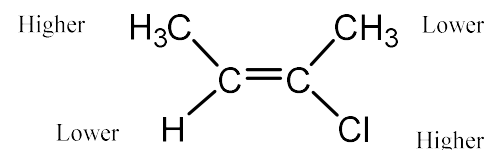
Z configuration



E configuration

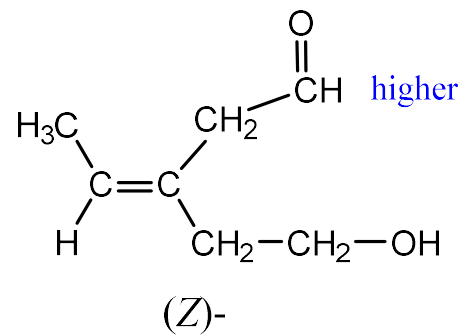
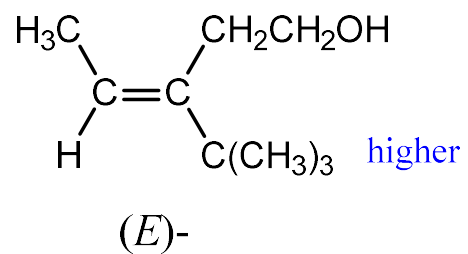
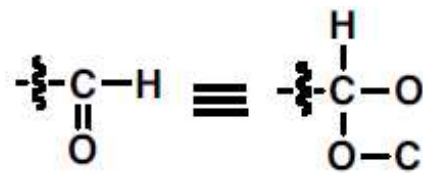
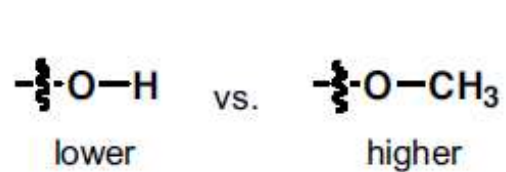


(Z)- 3-methyl-2-pentene



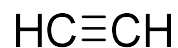
(E)- 2-chloro-2-butene

Multiple bonded atoms are equivalent to the same number of single bonded atoms.



Nomenclature of ALKYNES

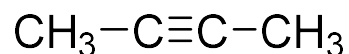
- -yne suffix (with number)



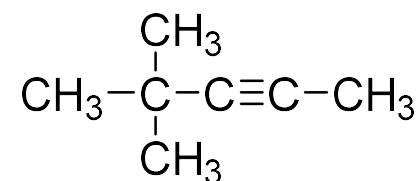
Ethyne



Propyne



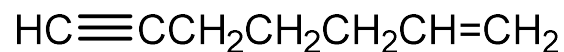
2-Butyne



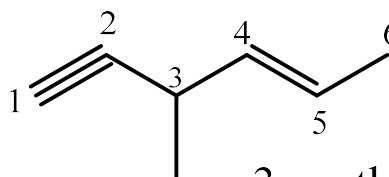
4,4-Dimethyl-2-pentyne

Enynes: containing both double and triple bond.

- suffix is -en-yne.
- numbering is from the end closer to a multiple bond.
- When there is equal chance in numbering, double bond receives lower number than triple bond.



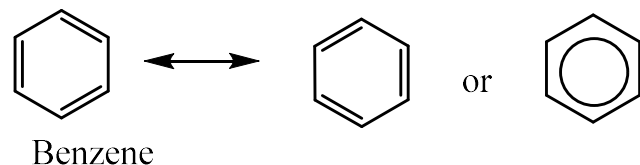
1-Hepten-6-yne



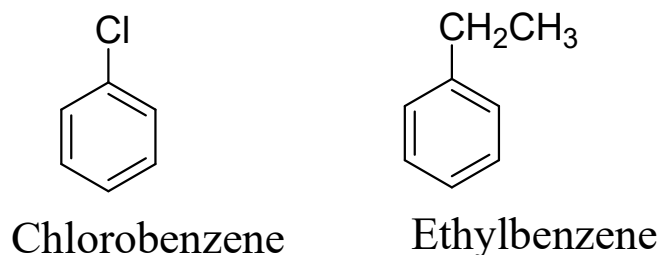
3-methyl-4-hexene-1-yne

Nomenclature of Aromatic Hydrocarbons

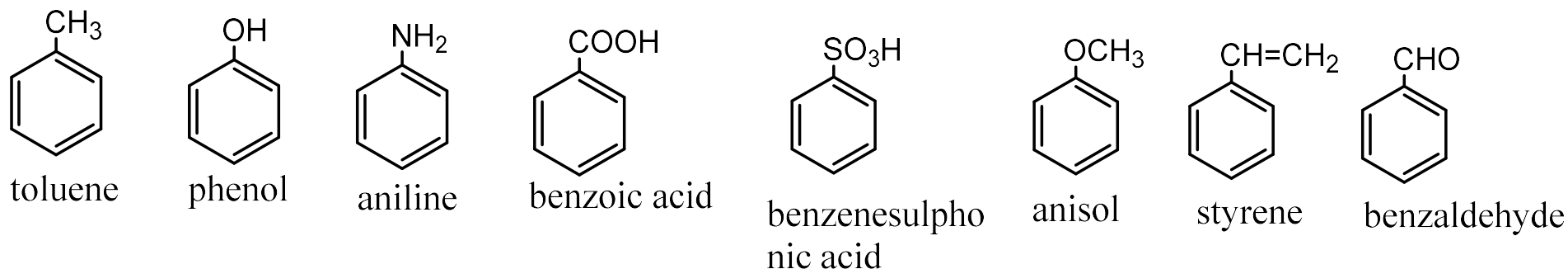
Aromatic hydrocarbon includes benzene and its derivative.



Monosubstituted benzenes are named as “substituent benzene”

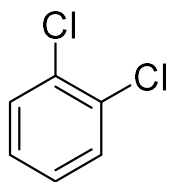
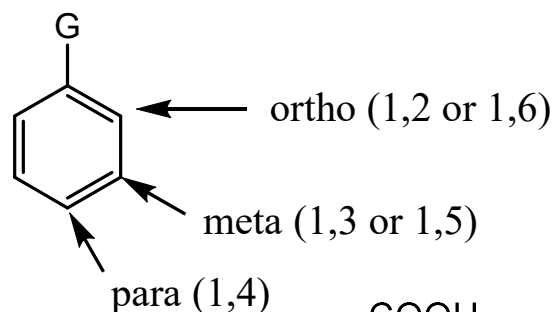


Common names of special monosubstituted benzenes that are also used in IUPAC system

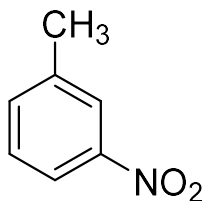


Disubstituted benzene

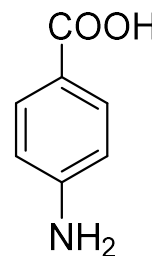
- Ortho-Meta-Para Designation
- Numbering approach



ortho-dichlorobenzene
(1,2-dichlorobenzene)



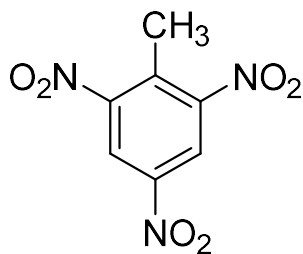
meta-nitrotoluene
(3-nitrotoluene)



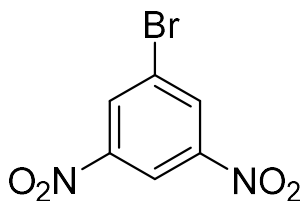
para-aminobenzoic acid
(4-aminobenzoic acid)

Polysubstituted benzene

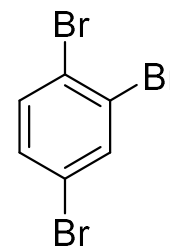
The benzene ring is numbered so as to give the lowest possible number to the substituent.



2,4,6-trinitrotoluene
(TNT)



3,5-dinitrobromobenzene



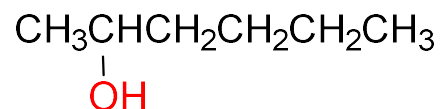
1,2,4-tribromobenzene

Nomenclature of ALCOHOL

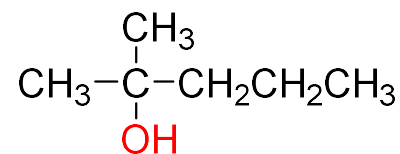
- -ol suffix with number designation
- OH group takes priority (even over -ene or -yne)
- the direction of numbering gives it the lowest possible number



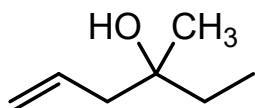
Ethanol



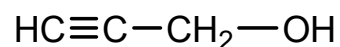
2-Hexanol



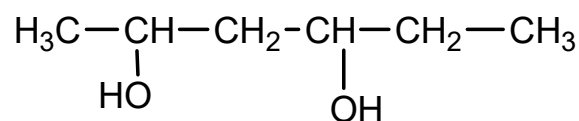
2-Methyl-2-pentanol



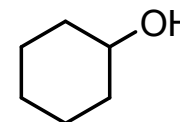
3-methyl-5-hexen-3-ol



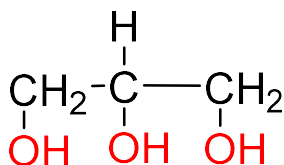
2-propynol



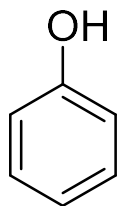
2,4-hexanediol
(Hexane-2,4-diol)



cyclohexanol



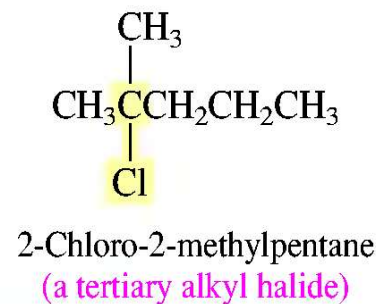
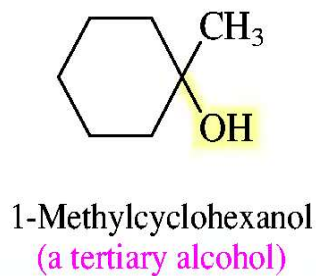
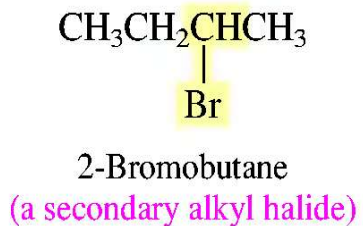
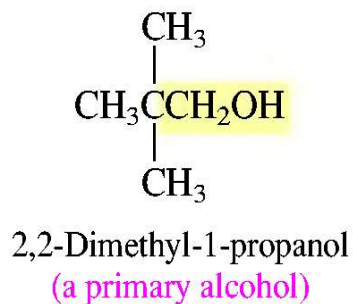
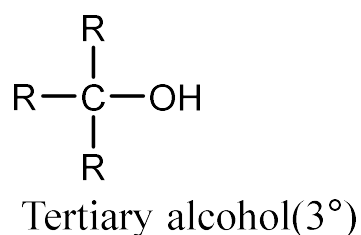
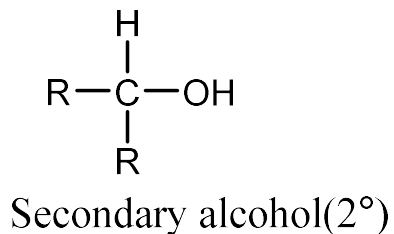
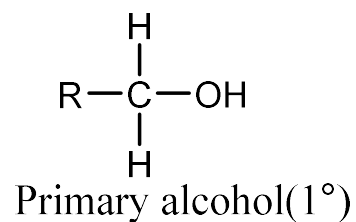
Propane-1,2,3-triol
(Glycerin)



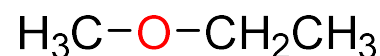
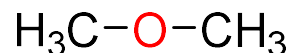
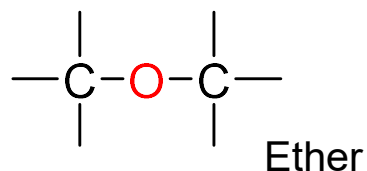
Phenol

CLASSES OF ALCOHOLS

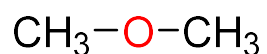
Alcohols are classified as primary, secondary, or tertiary according to the degree of substitution of the carbon that bears the functional group.



Nomenclature of Ethers and Epoxides

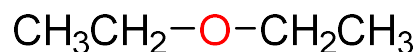


Naming: List the two alkyl groups in the general structure ROR' in alphabetical order as separate words, and then adding the word ether at the end.



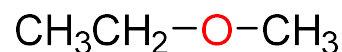
Dimethyl ether

(Methoxymethane)



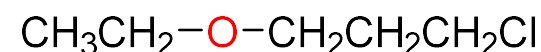
Diethyl ether

(Ethoxyethane)



Ethyl methyl ether

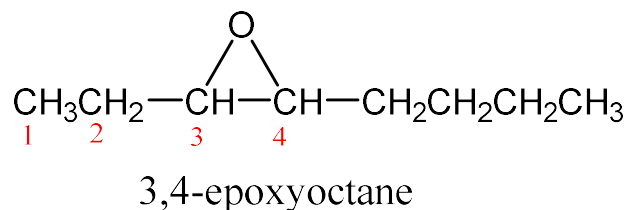
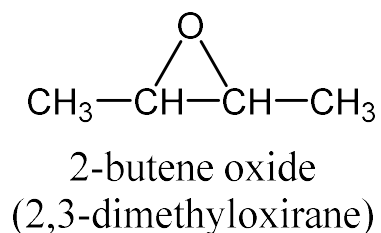
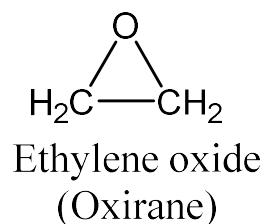
(Methoxyethane)



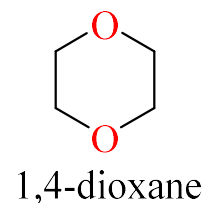
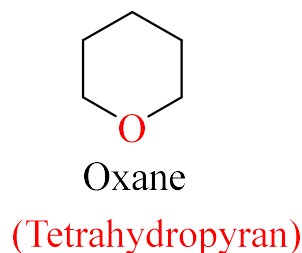
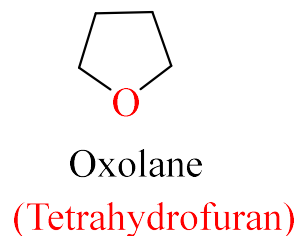
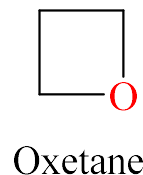
3-Chloropropyl ethyl ether

(1-Chloro-3-ethoxypropane)

Epoxide (Oxiranes) are cyclic ethers with three membered ring containing one oxygen



Cyclic Ethers

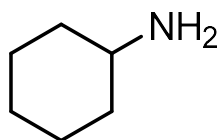


Nomenclature of Amines $R-NH_2$

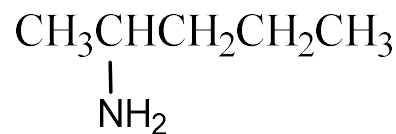
- Name either as *alkylamines* or as *alkanamines*
- When named as alkylamines, the ending **-amine** is added to the name of the alkyl group that bears the nitrogen.
- When named as alkanamines, the alkyl group is named as an alkane and the -e ending replaced by -amine.



Ethylamine
(Ethanamine)



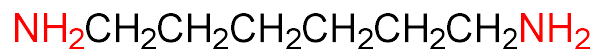
cyclohexanamine



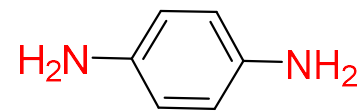
1-Methylbutylamine
(2-pentanamine)



1,2-Propanediamine



1,6-Hexanediamine



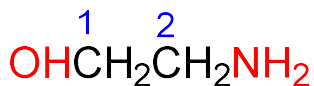
1,4-Benzenediamine

When two or more functional groups found in a molecule, the parent hydrocarbon is determined by priority order:

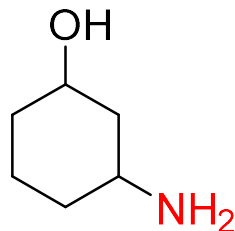
Priority order from highest to lowest:

Carboxylic acid > ester > amide > nitrile > aldehyde > ketone > alcohol > amine > alkene > alkyne

The amine group is named as a substituent when the amine group takes low priority

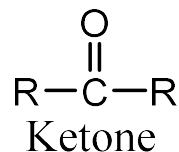
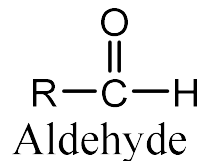


2-Aminoethanol



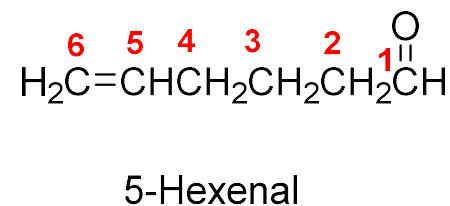
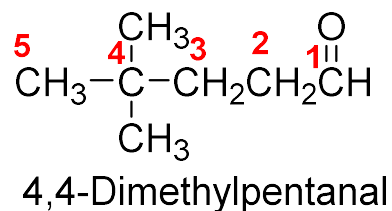
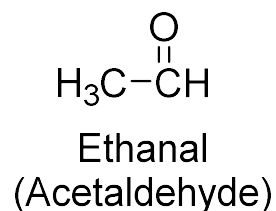
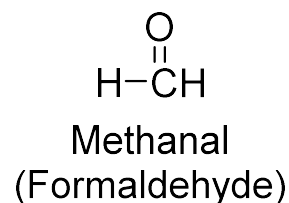
3-aminocyclohexanol

Nomenclature of Aldehydes and Ketones

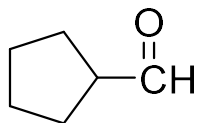


Aldehyde

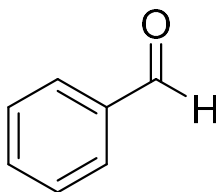
- The *-e* ending of the corresponding alkane name is replaced by *-al*
- The aldehyde group takes C-1 (not necessary to indicate the position of the aldehyde).



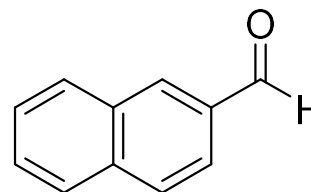
When a formyl group ($-\text{CH}=\text{O}$) is attached to a ring, the ring name is followed by the suffix -carbaldehyde.



Cyclopentanecarbaldehyde



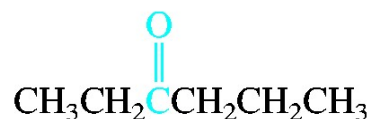
Benzenecarbaldehyde
(Benzaldehyde)



2-Naphthalenecarbaldehyde

Ketone

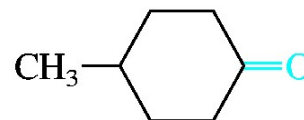
- -one suffix
- The carbonyl carbon of a cyclic ketone is C-1 and the number does not appear in the name.



3-Hexanone



4-Methyl-2-pentanone

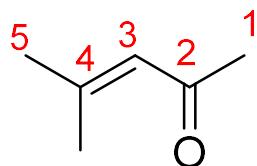


4-Methylcyclohexanone

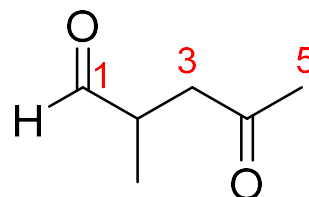
Priority order :

Carboxylic acid > ester > amide > nitrile > aldehyde > ketone > alcohol > amine > alkene > alkyne

The carbonyl oxygen of the ketone is considered an oxo-substituent in the presence of higher priority functional group.

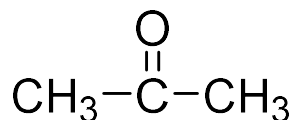


4-Methyl-3-penten-2-one

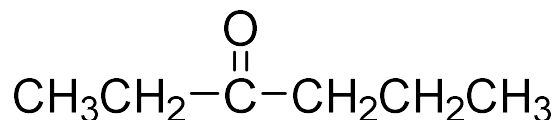


2-Methyl-4-oxopentanal

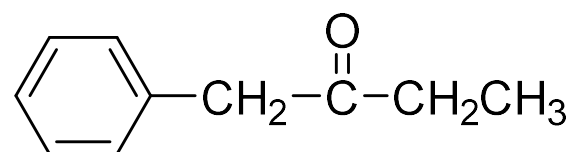
Ketones also named by functional class nomenclature.



Dimethyl ketone
(Acetone)

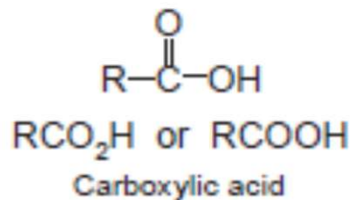
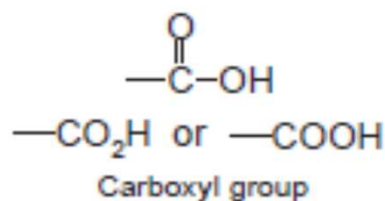


Ethyl propyl ketone

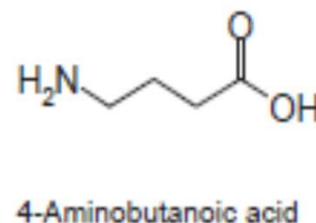
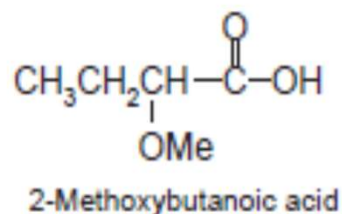
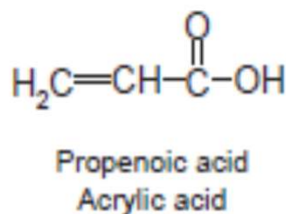
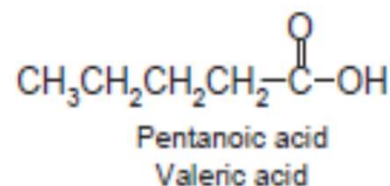
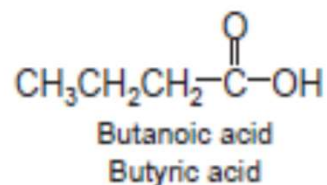
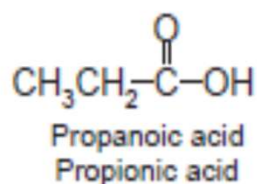


Benzyl ethyl ketone

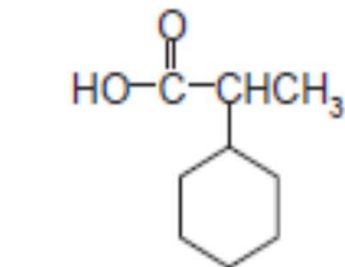
Nomenclature of Carboxylic Acids and their Derivatives



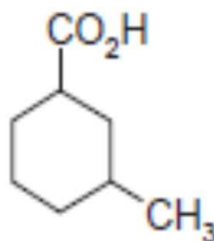
-oic acid suffix



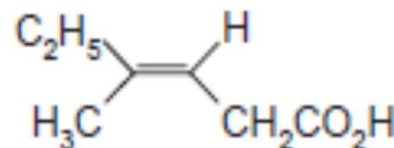
Cycloalkanes with carboxyl substituents are named as cycloalkanecarboxylic acids. Unsaturated acids are named using the name of the alkene with *-e* replaced with *-oic acid*.



2-Cyclohexylpropanoic acid

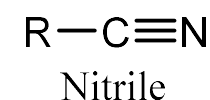
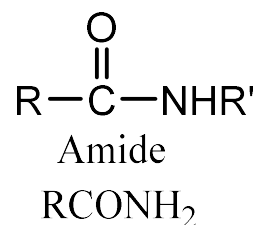
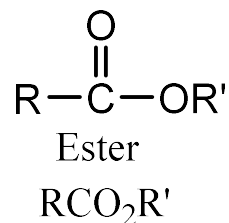
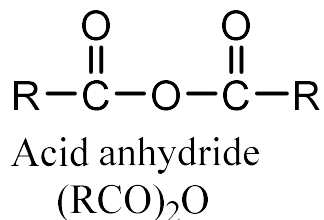
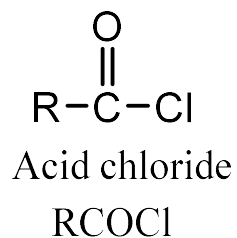


3-Methylcyclohexanecarboxylic acid

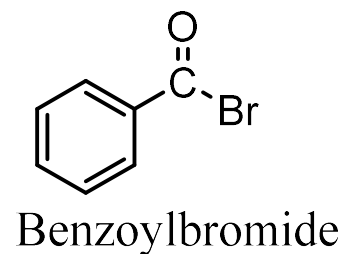
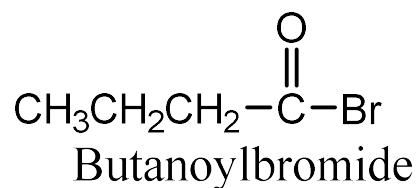
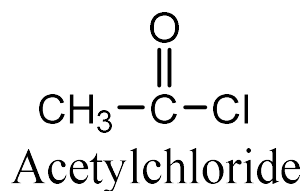
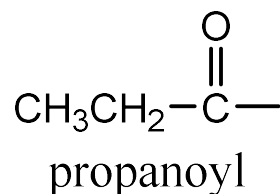
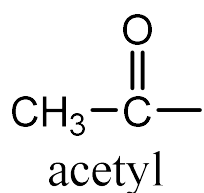
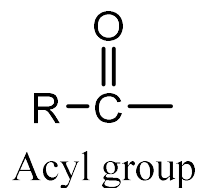


(E)-4-Methyl-3-hexenoic acid

Carboxylic Acids Derivatives

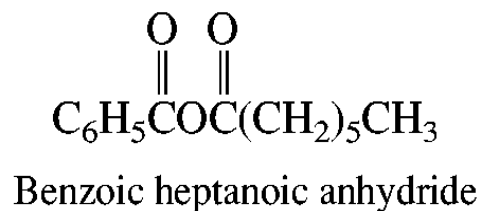
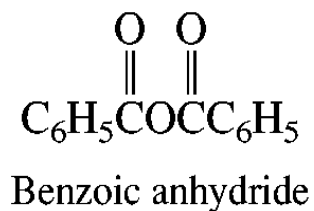
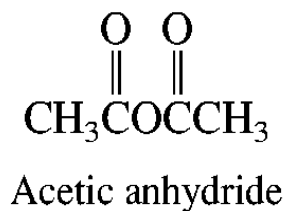


- Acyl groups are named by replacing the **-ic acid** ending of the corresponding carboxylic acid by **-yl**.
- Acyl halides are named by placing the name of the appropriate halide after that of the acyl group.

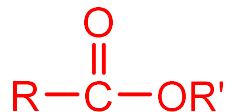


acid anhydrides

replace **acid** by **anhydride**.

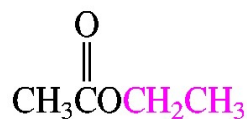


Esters

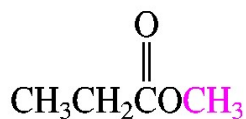


-named as alkyl alkanoate. R' cited first followed by the acyl portion.

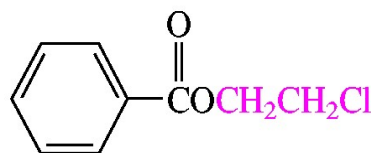
The acyl portion is named by substituting the suffix **-ate** for the **-ic acid** ending of the corresponding acid.



Ethyl acetate



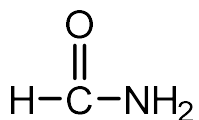
Methyl propanoate



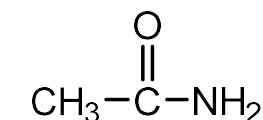
2-Chloroethyl benzoate

Amides

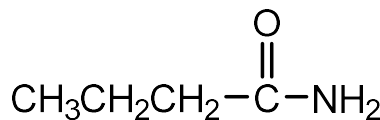
-replacing the suffix **-oic acid** or **-ic acid** by **-amide**



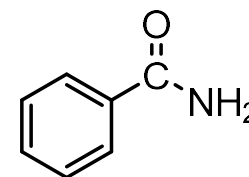
Formamide



Acetamide



Butanamide

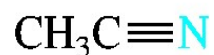


Benzamide

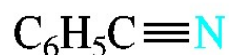
Nitriles $R-C\equiv N$

-add the suffix **-nitrile** to the name of the parent hydrocarbon chain that includes the carbon of the cyano group.

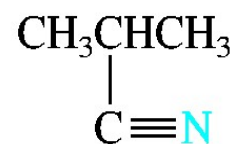
-by replacing the -ic acid or -oic acid ending of the corresponding carboxylic acid with *-onitrile*.



Ethanenitrile
(acetonitrile)



Benzonitrile



2-Methylpropanenitrile
(isopropyl cyanide)

Physical Properties

Alkanes

- Methane, ethane, propane, and butane are gases at room temperature.
- The unbranched alkanes pentane (C_5H_{12}) through heptadecane ($\text{C}_{17}\text{H}_{36}$) are liquids.
- Higher homologs are solids.
- The boiling points of unbranched alkanes increase with the number of carbons.

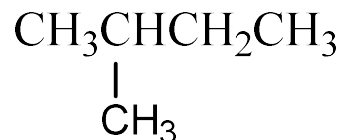
Name	Number of carbons	Molecular formula	Condensed structure	bp ($^{\circ}\text{C}$)
Methane	1	CH_4	CH_4	-164
Ethane	2	C_2H_6	CH_3CH_3	-88.6
Propane	3	C_3H_8	$\text{CH}_3\text{CH}_2\text{CH}_3$	-42.1
Butane	4	C_4H_{10}	$\text{CH}_3(\text{CH}_2)_2\text{CH}_3$	-0.60
Pentane	5	C_5H_{12}	$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$	36.1
Hexane	6	C_6H_{14}	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	68.9
Heptane	7	C_7H_{16}	$\text{CH}_3(\text{CH}_2)_5\text{CH}_3$	98.4
Octane	8	C_8H_{18}	$\text{CH}_3(\text{CH}_2)_6\text{CH}_3$	125.7

- Branched alkanes have lower boiling points than their unbranched isomers.



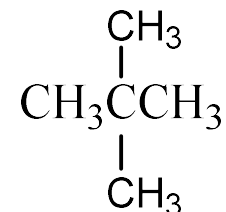
b.pt 36°C

pentane



b.pt 28°C

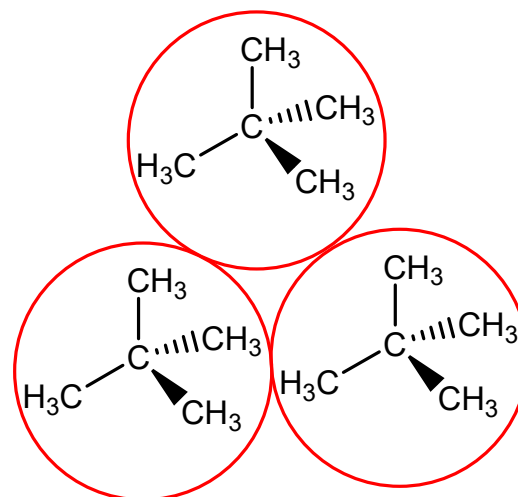
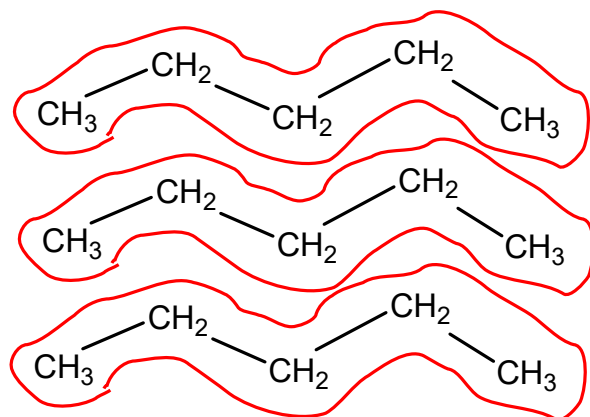
methylbutane



b.pt 9°C

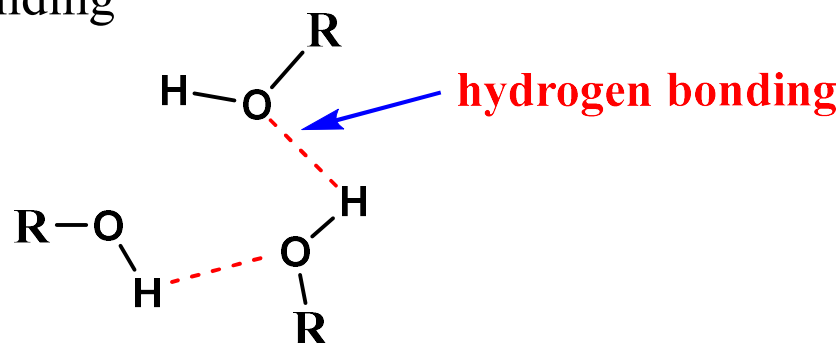
2,2-dimethylpropane

- Pentane molecules have more van der Waals attraction than the other.
 - Straight chain pentane molecules have more surface area available for contact between them than the spherical 2,2-dimethylpropane.



Alcohols

- Form hydrogen bonding



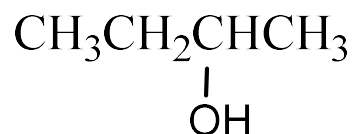
- The boiling points of alcohols are in general much higher than comparably sized hydrocarbon, alkyl halide, ethers and carbonyl compounds.

	Ethanol CH_3CH_2OH	Dimethylether CH_3-O-CH_3	Propane $CH_3CH_2CH_3$
Mol wt	46	46	44
b.pt	+ 78°C	-24°C	-42°C

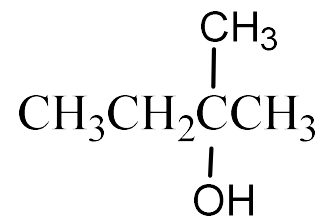
- Branching tend to decrease the boiling point since a more symmetrical molecules have less Dispersion forces



b.pt = 118°C

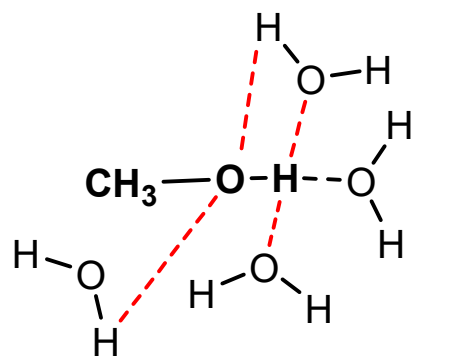


b.p t= 100°C

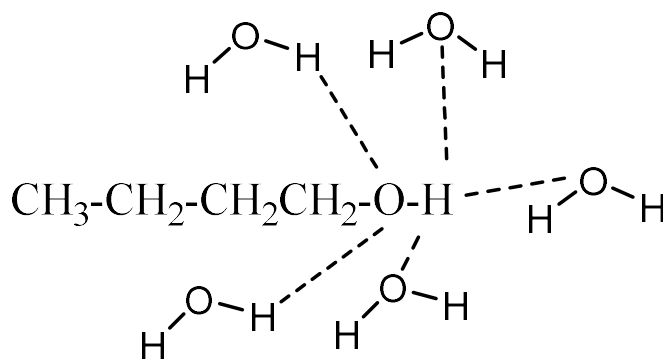


b.pt = 83°C

- Low molecular weight alcohols (methyl, ethyl, n-propyl, and isopropyl alcohol) are soluble in water in all proportions.
- With smaller molecules water can more easily surround the alcohol molecules effectively separating them from one another (solvation).

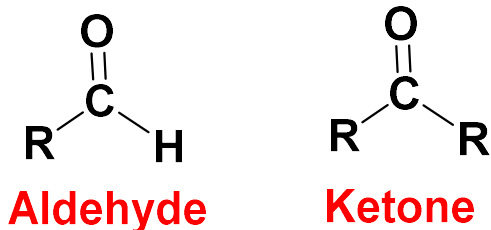


solvation of methanol

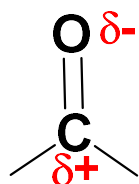


solvation of 1-butanol

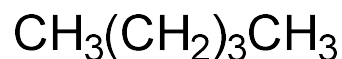
Carbonyl compounds



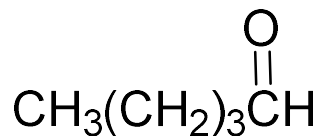
- Oxygen is more electronegative than carbon. Therefore the electrons in the C=O bond are attracted to oxygen, producing a highly polarized bond.



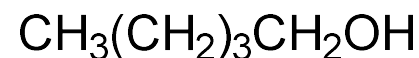
- Carbonyl compounds boil at higher temperature than hydrocarbons, but lower temperature than alcohols.



Pentane
b.pt 36 °C



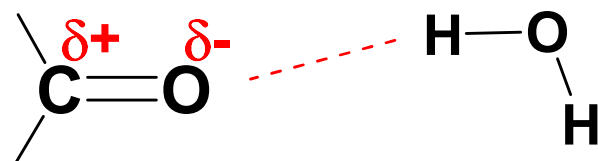
Pentanal
b.pt 75 °C



Pentanol
b.pt 118 °C

- Carbonyl compounds have permanently polarized C=O bond and thus have strong tendency to associate (dipole-dipole interaction)
- Dipole-dipole interactions are stronger than van der Waals force but not as strong as hydrogen bond

- Carbonyl compounds don't form hydrogen bond with one another.
- Low molecular weight carbonyls are soluble in water, form hydrogen bond with O-H and N-H



Exercise

Why carboxylic acids have high boiling points than those alcohol with comparable molecular weight? For example, acetic acid and propyl alcohol, which have the same formula weight (60), boils at 118 °C and 97 °C, respectively.

	<chem>CH3CH2CH2OH</chem>	<chem>CH3C(=O)OH</chem>
Boiling point	97 °C	118°C